

Asymptotic analysis of a trigonometric reformulation associated with Fermat's equation

Two parameter regimes with Lean 4 verification

ChatGPT codex

Abstract

This manuscript develops a partial solution framework for Fermat's Last Theorem through a trigonometric reformulation. A hypothetical positive solution of $x^n + y^n = z^n$ determines an acute triangle and an associated right triangle, yielding $\tan^2 \theta = (\sin \alpha / \sin \beta)^n$. The induced real equation is first classified in two regimes. For $0 < a < b < 1$, with $b = 1$ also covered, every root sequence converges to zero. For general $b > a$, every interior nonsingular accumulation point with positive sine ratio lies on the level set $\sin(at)/\sin(bt) = 1$. Combining this rigidity with the Fermat fact that at and bt are acute angles gives a stronger unified conclusion: for fixed frequencies, the acute branch can only collapse to zero. Lean 4 verifies all three statements. This remains a partial solution because it constrains fixed-frequency trigonometric families rather than excluding every individual integer solution.

Meaning of Partial Solution

The phrase partial solution is used here in a precise restricted sense. The Fermat equation is converted into an exact necessary trigonometric identity, and the possible asymptotic behavior of that identity is resolved in two substantial parameter regimes. This eliminates stable interior root behavior in the nonoscillatory case and identifies the only possible regular interior limits in the general case. What remains unresolved is whether the angle parameters arising from one fixed integer triple must enter either limiting mechanism in a way that produces a contradiction.

Accordingly, the manuscript supplies a partial analytic solution of the trigonometric problem generated by Fermat's equation. It does not supply a partial proof of the number-theoretic theorem in the usual formal sense: no exponent $n > 2$ is proved impossible for integer x, y , and z solely from these results. This distinction allows the geometric progress to be stated positively without overstating its logical consequence.

Problem Setting

Fix real parameters a and b in the nonoscillatory range $0 < a < b \leq 1$. For $0 < t < \pi/2$, define

$$r(t) = \frac{\sin(at)}{\sin(bt)}, \quad 0 < a < b \leq 1,$$

We consider a sequence of positive roots t_n satisfying

$$r(t_n)^n = \tan^2(t_n), \quad 0 < t_n < \frac{\pi}{2}.$$

The restriction $b \leq 1$ keeps both at and bt inside the first quadrant. It prevents the denominator from oscillating or acquiring internal zeros. The main conclusion is

$$\lim_{n \rightarrow \infty} t_n = 0.$$

Relation to Fermat's Last Theorem

Let x , y , and z be positive numbers satisfying Fermat's equation

$$x^n + y^n = z^n, \quad n > 2,$$

After interchanging x and y if necessary, assume $x < y < z$. The equation implies $x^2 + y^2 > z^2$, so x , y , and z form an acute triangle. Let α , β , and γ be the angles opposite x , y , and z . The sine rule gives

$$\frac{x}{y} = \frac{\sin \alpha}{\sin \beta},$$

The powered lengths $x^{n/2}$, $y^{n/2}$, and $z^{n/2}$ form a right triangle because their squares satisfy the Fermat equation. Define θ as the acute angle whose opposite and adjacent legs are $x^{n/2}$ and $y^{n/2}$. Then

$$\tan \theta = \frac{x^{n/2}}{y^{n/2}} = \left(\frac{x}{y} \right)^{n/2},$$

Squaring and substituting the sine-rule ratio yields the exact identity

$$\tan^2 \theta = \left(\frac{x}{y} \right)^n = \left(\frac{\sin \alpha}{\sin \beta} \right)^n.$$

This identity is a quasi-solution only in the sense of a geometric reduction: it converts a hypothetical Fermat solution into a necessary relation between the angles of two linked triangles. It does not contradict the existence of integer x , y , and z . To compare it with the root equation below, one may write $a = \alpha/\theta$, $b = \beta/\theta$, and $t = \theta$. However, the required condition $b \leq 1$ would mean $\beta \leq \theta$, and the Fermat construction does not establish that inequality. Even in the arbitrary-frequency regime, proving that an accumulation point lies on $\sin(at)/\sin(bt) = 1$ supplies only another necessary trigonometric condition. Such level-set points can exist for real angles and contain no contradiction with integrality. It is therefore not a valid solution of Fermat's Last Theorem.

What the Reduction Reveals About Fermat's Equation

The construction is informative because it separates a hypothetical Fermat solution into two geometries. The original lengths x , y , and z determine an acute triangle, while the powered lengths determine a right triangle. The same ratio x/y is encoded once by the sine rule and again by a tangent. Thus the exponent n no longer appears only in an integer equation; it becomes the power that links two angle systems.

This change of viewpoint exposes an amplification effect. If $\sin \alpha / \sin \beta$ is strictly below one and remains uniformly separated from one, its n th power becomes very small. The right-triangle angle θ must then be small because $\tan^2 \theta$ has the same value. In the nonoscillatory model this mechanism is rigorous: all roots move toward zero. Translated back to the triangle picture, large exponents force a strong angular imbalance rather than producing a stable interior configuration.

The arbitrary-frequency theorem supplies the complementary possibility. A sequence that does retain an interior nonsingular angular limit must make the underlying sine ratio approach one. It is therefore driven toward $\sin \alpha = \sin \beta$. For angles of one acute triangle, sine is strictly increasing, so this would imply $\alpha = \beta$ and hence $x = y$. This observation can be useful only when the normalised parameters remain in the same nonoscillatory branch and the limiting process is justified; the general Fermat construction does not provide those facts automatically.

For a fixed pair $a < b$, however, the Fermat construction supplies exactly the branch information needed at every root: $at = \alpha$ and $bt = \beta$ are acute, so $0 < at < bt < \pi/2$. Consequently $\sin(at) < \sin(bt)$ at every positive point of the branch. The level-set alternative from the general-frequency theorem is therefore impossible there. Poles are also impossible because $\sin(bt)$ is positive. This leaves $t = 0$ as the only possible limiting location for a fixed-frequency root family.

Most importantly, the analysis identifies the exact missing bridge. Fermat's Last Theorem concerns one fixed integer exponent and exact positive integers, whereas root collapse is an asymptotic statement across growing exponents and the level-set theorem concerns limits of real roots. Neither converts an individual integer solution into a forbidden equality. A proof of Fermat's Last Theorem would still need an arithmetic argument showing that the necessary angular configuration cannot arise from integer side lengths. The trigonometric results diagnose rigidity and possible limiting behavior, but do not supply that arithmetic obstruction.

Partial Result One Nonoscillatory Collapse

Theorem 1 Root collapse below the unit frequency boundary

Let $0 < a < b < 1$. Suppose t_n lies in $(0, \pi/2)$ for every natural number n and satisfies $r(t_n)^n = \tan^2(t_n)$. Then t_n tends to zero as n tends to infinity. The same proof includes the boundary value $b = 1$.

Proof

For any t in $(0, \pi/2]$, the parameter assumptions give

$$0 < at < bt \leq \frac{\pi}{2},$$

The sine function is strictly increasing on the first quadrant. Therefore

$$0 < \sin(at) < \sin(bt), \quad 0 < r(t) < 1.$$

Fix $\delta > 0$ with $\delta < \pi/2$. The ratio r is continuous on the compact interval $[\delta, \pi/2]$. Its denominator is strictly positive there, so the extreme value theorem gives a maximum

$$M_\delta = \max_{\delta \leq t \leq \pi/2} r(t) < 1.$$

The strict inequality follows because every value of r on this interval is smaller than one. It follows uniformly that

$$0 < r(t)^n \leq M_\delta^n \longrightarrow 0.$$

At the same time, tangent is positive and increasing on the first quadrant, hence

$$\tan^2(t) \geq \tan^2(\delta) > 0.$$

Choose N so large that $M_\delta^n < \tan^2(\delta)$ whenever $n \geq N$. For such n , every t in $[\delta, \pi/2]$ satisfies

$$r(t)^n < \tan^2(t) \quad \text{for all } t \in [\delta, \pi/2)$$

No root t_n can therefore remain in $[\delta, \pi/2)$ once n is sufficiently large. Since δ was arbitrary, t_n eventually lies below every positive δ . This is exactly the statement that t_n tends to zero.

Existence and Local Scale

The theorem is conditional on a sequence of roots, but roots do exist for each positive exponent in this regime. As t approaches zero from the right, $r(t)^n$ tends to $(a/b)^n$, which is positive, while $\tan^2(t)$ tends to zero. Near $\pi/2$, the ratio remains bounded and $\tan^2(t)$ diverges. Continuity therefore supplies at least one intersection.

The Taylor expansions at the origin give

$$r(t) = \frac{a}{b} \left(1 + \frac{b^2 - a^2}{6} t^2 + O(t^4) \right),$$

$$\tan^2 t = t^2 + O(t^4),$$

Substitution into the root equation suggests, and a refined estimate confirms, the exponential scale

$$t_n \sim \left(\frac{a}{b} \right)^{n/2}.$$

This estimate explains the geometry of the proof: the roots enter an exponentially shrinking neighborhood of the origin as the exponent grows.

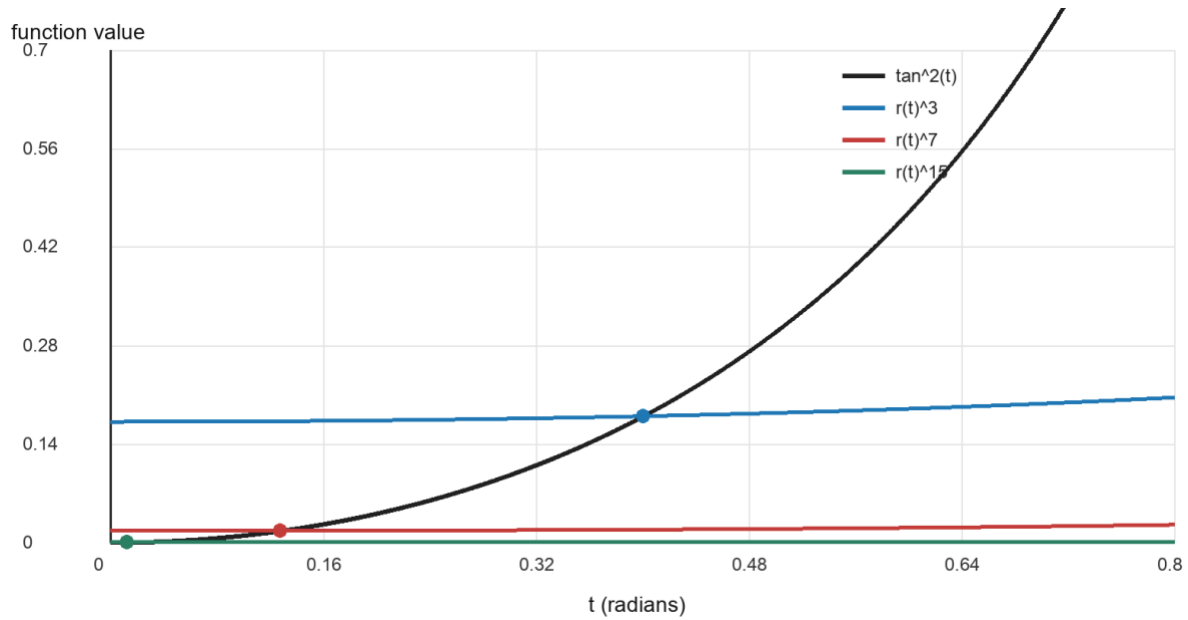


Figure 1. Intersections for $a = 0.5$ and $b = 0.9$. The marked roots move toward zero as n increases.

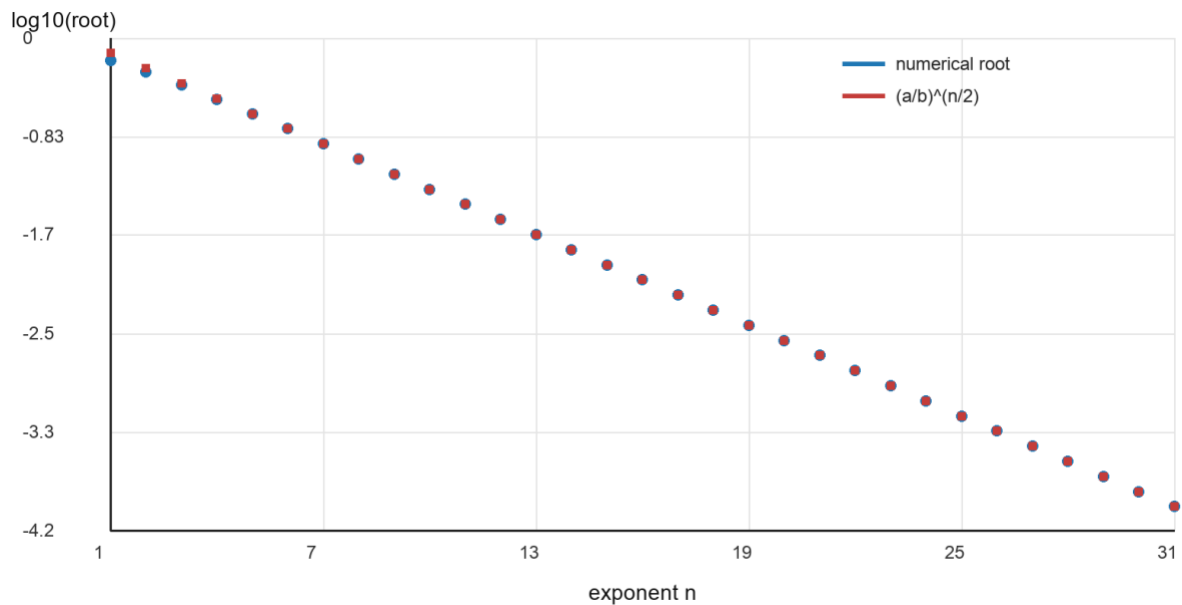


Figure 2. The numerical roots follow the predicted exponential scale $(a/b)^{n/2}$.

Partial Result Two General Frequencies

The first partial result does not extend as a global collapse theorem to every $b > a$. The general regime has a different, equally important conclusion: regular interior accumulation can occur only on a sharply specified level set.

When $b > a$ without the restriction $b \leq 1$, the denominator $\sin(bt)$ may change sign or vanish in $(0, \pi/2)$. The interval splits into sign components separated by poles. Roots with odd exponent can occur only where $\sin(at)$ and $\sin(bt)$ have the same sign, because $\tan^2(t)$ is positive.

Let t_n be positive roots and suppose a subsequence converges to an interior point t^* with $\sin(bt^*) \neq 0$. Assume the corresponding sine ratios are positive. Continuity makes $\tan^2(t_n)$ converge to the finite positive number $\tan^2(t^*)$. If the limiting ratio were below one, its n th powers would decay to zero; if it were above one, they would grow without bound. Both alternatives contradict the root equation. Consequently every such interior nonsingular accumulation point satisfies

$$\frac{\sin(at_*)}{\sin(bt_*)} = 1,$$

This conclusion does not say that every root converges to such a point. Other branches may approach zero, a pole of the ratio, or the endpoint $\pi/2$. It also does not assert that every level-set point is approached by roots; the condition is necessary, not sufficient.

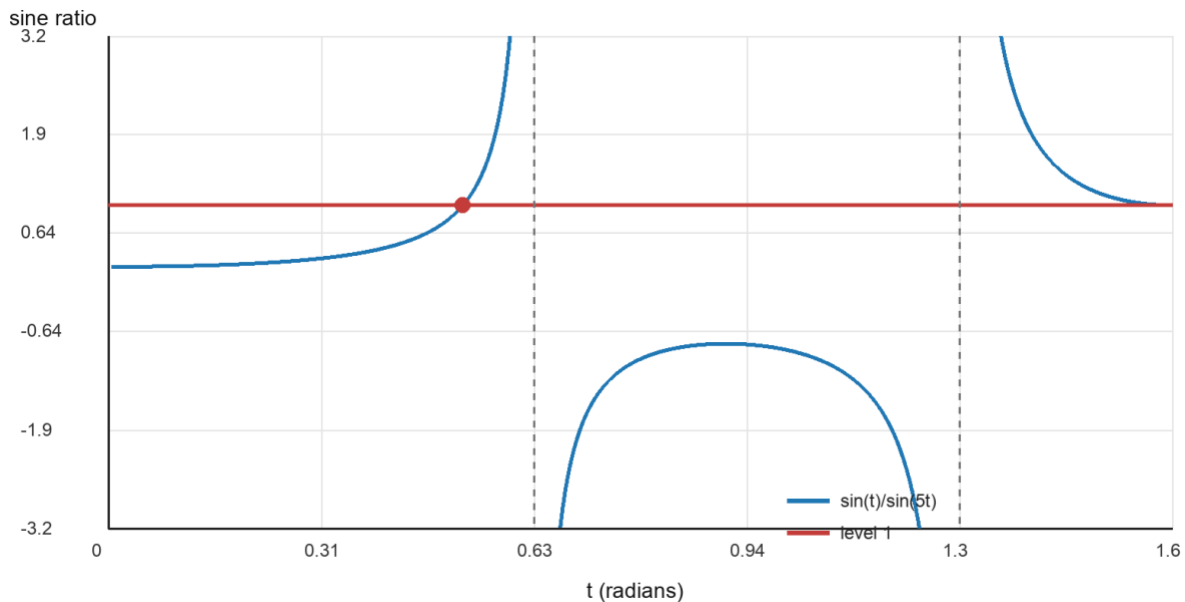


Figure 3. In the oscillatory example $a = 1$ and $b = 5$, poles split the interval into branches; red points mark nonsingular intersections with the level one.

Unified Fixed Frequency Consequence

Now impose the geometric information available in the Fermat construction. For fixed $0 < a < b$, suppose every root t_n satisfies

$$0 < a t_n < b t_n < \pi/2.$$

The denominator is then always positive and no root lies on an oscillatory or singular branch. If a positive interior accumulation point t^* existed, Partial Result Two would force $\sin(at^*)/\sin(bt^*) = 1$. Strict monotonicity of sine on the first quadrant instead gives $\sin(at^*) < \sin(bt^*)$, a contradiction. Compactness and the same uniform power estimate used in Partial Result One therefore yield

$$t_n \rightarrow 0.$$

This removes the global restriction $b \leq 1$. The real hypothesis is rootwise acuteness, which is already present when at_n and bt_n are the two ordered acute angles α_n and β_n of the Fermat triangle. Thus, for fixed frequencies, collapse to the origin is the only asymptotic branch compatible with the Fermat geometry.

The qualifier fixed frequencies remains essential. Across hypothetical Fermat triples with changing exponent, the normalized quantities $a_n = \alpha_n/\theta_n$ and $b_n = \beta_n/\theta_n$ may vary. The theorem does not prove that they can be replaced by one fixed pair or controlled uniformly. This is the remaining logical gap between the collapse theorem and a proof of Fermat's Last Theorem.

Lean Verification

The theorem was formalized in Lean 4 with Mathlib. The formal proof follows the compactness argument directly. It proves positivity and strict boundedness of the sine ratio, obtains a maximum on each closed interval away from zero, invokes geometric decay of powers below one, and derives convergence through the order characterization of limits.

```
theorem roots_tend_to_zero
  {a b : ℝ} (ha : 0 < a) (hab : a < b) (hb : b ≤ 1)
  (t : ℕ → ℝ)
  (ht : ∀ n, t n ∈ Set.Ioo (0 : ℝ) (Real.pi / 2))
  (hroot : ∀ n,
    (Real.sin (a * t n) / Real.sin (b * t n)) ^ n =
      Real.tan (t n) ^ 2) :
  Filter.Tendsto t Filter.atTop (nhds 0)
```

The checked source is `RootCollapse.lean` in the accompanying `NivenProof` project. The file compiles without placeholders or admitted results. The formal statement covers all natural exponents; oddness is unnecessary because the sine ratio is positive throughout the stated parameter range.

The arbitrary-frequency accumulation statement is formalized separately. It assumes convergence to an interior point, a nonzero limiting denominator, positivity of the ratios, and the root equation. It then proves the level-set conclusion.

```
theorem accumulation_point_on_level_set
  {a b tStar : ℝ} (t : ℕ → ℝ)
  (ht : Tendsto t atTop (ℳ tStar))
  (htStar : tStar ∈ Ioo (0 : ℝ) (Real.pi / 2))
  (hdenom : Real.sin (b * tStar) ≠ 0)
  (hrpos : ∀ n, 0 < Real.sin (a * t n) / Real.sin (b * t n))
  (hroot : ∀ n, (Real.sin (a * t n) / Real.sin (b * t n)) ^ n
    = Real.tan (t n) ^ 2) :
  Real.sin (a * tStar) / Real.sin (b * tStar) = 1
```

The fixed-frequency Fermat consequence is captured by a third theorem. It replaces the global condition $b \leq 1$ with the rootwise acute-branch condition $b \cdot t_n < \pi/2$.

```

theorem roots_tend_to_zero_of_acute_branch
  {a b : ℝ} (ha : 0 < a) (hab : a < b)
  (t : ℕ → ℝ)
  (ht : ∀ n, t n ∈ Ioo (0 : ℝ) (Real.pi / 2))
  (hacute : ∀ n, b * t n < Real.pi / 2)
  (hroot : ∀ n, (Real.sin (a * t n) / Real.sin (b * t n)) ^ n
    = Real.tan (t n) ^ 2) :
  Tendsto t atTop (ℕ 0)

```

Scope of the Partial Solution

The condition $b \leq 1$ is essential to the global collapse theorem. If $b > 1$, the argument bt may leave the first quadrant, $\sin(bt)$ may vanish inside the interval, and the ratio can change sign or develop poles. Root branches may then approach internal solutions of $\sin(at) = \sin(bt)$, a pole, or the endpoint $\pi/2$ rather than zero.

Together, the results form the partial solution claimed in the title. When $0 < a < b < 1$, roots collapse to $t = 0$. For arbitrary $b > a$, a regular interior limit must lie on the level set $\sin(at)/\sin(bt) = 1$. When the Fermat acute-angle condition is restored and the frequencies are fixed, that level set is inaccessible and the general case also collapses to zero. These conclusions still do not establish Fermat's Last Theorem because fixed frequencies are not known to persist across hypothetical triples with varying exponent.

Conclusion

The trigonometric analysis gives a three-step partial solution to the equation induced by a hypothetical Fermat triple. First, the nonoscillatory regime collapses to zero. Second, general regular interior limits are confined to the level set $\sin(at)/\sin(bt) = 1$. Third, because the Fermat quantities at and bt are ordered acute angles, that level set is impossible for fixed frequencies, so only collapse to zero remains. Lean 4 checks all three statements. A complete solution of Fermat's Last Theorem would still have to control the varying normalized frequencies or derive an arithmetic contradiction for each fixed exponent and positive integer triple.